Journal of Affective Disorders 399 (2026) 121056

Contents lists available at ScienceDirect

Journal of Affective Disorders

journal homepage: www.elsevier.com/locate/jad

Research paper

A preliminary investigation of a Geriatric Depression Scale, Dementia version (GDS-D)

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A B S T R A C T

Background: Depression is a frequent comorbidity in dementia; however, diagnosis is complicated by overlapping cognitive and behavioural symptoms. The Geriatric Depression Scale (GDS) is a widely used screening tool to probe for depression, comprising 30 items in its full version (GDS-30) but also existing as a range of shorter versions (GDS-15, GDS-10, GDS-8, GDS-4). All versions contain items that may be confounded in dementia patients such as probes for cognitive symptoms, apathy, and negative feelings about the future. This study aimed to develop and validate a 10-item version (GDS—D) optimised for diagnosing depression in dementia by removing these confounds.

Methods: Data were retrospectively analysed in participants with Alzheimer's disease (AD, $n = 29$), frontotemporal dementia (FTLD, $n = 30$), or a depressive disorder ($n = 23$) who had completed the GDS-30 and undergone psychiatric assessment and application of the Diagnostic and Statistical Manual of Mental Disorders, fifth edition (DSM-5) criteria. Diagnostic accuracy of the GDS-D was compared to all previous GDS versions using sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV).

Results: The GDS-D had an accuracy of 92–97 % in diagnosing depression in a memory clinic setting and in dementia (AD and FTLD), which was superior to the GDS-30 and all briefer versions.

Conclusion: The GDS-D enhances depression screening in a memory clinic setting by excluding the confounding items of other GDS versions. In addition, it performs better at diagnosing depression in a dementia cohort. Future studies should validate its utility in larger, diverse dementia populations.

1. Introduction

Depression and degenerative dementias can share overlapping symptoms making differential diagnosis challenging (Goodarzi et al., 2017). The Geriatric Depression Scale (GDS) is a widely used screening tool for depression in older adults, designed to minimise the influence of somatic symptoms that can confound other depression scales in older adults (Burke et al., 2019). Though the GDS is more appropriate for older adults, it may be confounded by symptoms of dementia (de Boer et al., 2025; Flavell and Nestor, 2022). The GDS includes items related to cognitive symptoms, apathy, feelings of helplessness and pessimism about the future; all of which may occur in dementia patients independent of depression. These confounded items may lead to false positive diagnoses of depression in dementia cohorts. While a shorter version, the GDS-15, was created to target cohorts with dementia–it still includes several of these confounding items (Yesavage and Sheikh, 1986). Even briefer versions of the GDS (i.e., GDS-10, GDS-8 and GDS-4) have been proposed that exclude some of these questions, yet they have

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https://doi.org/10.1016/j.jad.2025.121056
Received 17 October 2025; Received in revised form 10 December 2025; Accepted 24 December 2025
Available online 28 December 2025
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been shown to be inaccurate for dementia cohorts (Goodarzi et al., 2017; Atchison et al., 2024). To address these limitations in existing GDS versions, this study aimed to develop and validate a modified version of the GDS for assessment of depression in dementia populations (the GDS—D) that minimised confounds by focusing particularly on items probing depressed mood and anhedonia.

2. Methods

2.1. Study design

Data from the Mater Memory and Cognitive Disorders Clinic and The University of Queensland were retrospectively reviewed consecutively between July 2018 and November 2025 for patients referred with suspected dementia who had completed the GDS-30 and had undergone a concurrent psychiatric assessment to determine the presence or absence of a depressive disorder (e.g., major depressive disorder, persistent depressive disorder, etc.) as defined by the Diagnostic and Statistical Manual of Mental Disorders, fifth edition (DSM-5) criteria (Association AP, 2013). All patients included had been assessed by a consultant psychiatrist (J.F., K.R., P.L.P.M., or P.E.M.). The study took place in Queensland, Australia.

2.2. Inclusion and exclusion criteria

Patients that had a final diagnosis of a depressive disorder, Alzheimer's disease (AD), or a clinical presentation of frontotemporal lobar degeneration (FTLD) were included. All patients in the dementia groups were required to have a Mini-Mental State Examination (MMSE) score of ≥15/30 for inclusion. All participants were required to have completed the GDS-30 and to have undergone the gold-standard assessment of a clinical interview conducted by a psychiatrist applying DSM-5 criteria for a depressive disorder.

2.3. Patients

A total of 82 patients were included in the study, comprising 30 people with FTLD, 29 with AD, and 23 with a depressive disorder. The FTLD cohort included patients diagnosed with behavioural variant frontotemporal dementia (bvFTD, $n = 12$), progressive aphasia (PPA, $n = 9$), progressive supranuclear palsy (PSP, $n = 6$), and FTD–motor neuron disease (FTD–MND, $n = 3$). Patients were diagnosed according to established research criteria (Gorno-Tempini et al., 2011; Rascovsky et al., 2011; Höglinger et al., 2017; Strong et al., 2017; Jack Jr. et al., 2024). The bvFTD cohort were all diagnosed with definite ($n = 2$) or probable ($n = 10$) bvFTD. The probable bvFTD cohort had follow-up of 2.8 ± 1.1 years (range: 2.0–5.2 years) to ensure diagnostic stability. The PPA cohort all had negative AD biomarkers on cerebrospinal fluid (CSF) analysis (Graff-Radford et al., 2021). The AD cohort all had positive AD biomarkers on CSF analysis.

Regarding diagnostic classification by the psychiatrists, there were 41 patients with a depressive disorder: including 23 with depression without dementia and 18 with comorbid depression and dementia ($n = 10$ FTLD and $n = 8$ AD). There were 41 euthymic patients without depression, including 20 with FTLD and 21 with AD. All patients except five (n = 1 euthymic FTD and $n = 4$ depression without dementia) had the GDS and psychiatric assessments administered on the same day; for the remainder, the GDS was administered within one month (21.2 ± 8.8 days; range: 8–30).

2.4. Data collection

Data were retrospectively collected from patients' electronic medical records at their initial clinic presentation. If the GDS was not recorded at that visit, data from the next documented appointment were used *if* concurrent with a psychiatric assessment. Demographic and cognitive data included age, sex, education level, Addenbrooke's Cognitive Examination third edition (ACE-III) (So et al., 2018), and Mini-Mental State Examination (MMSE) (Folstein et al., 1975) (Supplementary Table 1).

2.5. Statistical analysis

Statistical analyses were conducted using R statistical software (version 4.4.2) using the dplyr package (version 1.1.4) (Team RC, 2024). Test performance was determined by sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and diagnostic accuracy. These were calculated at the optimal cut-off points yielded by the highest Youden index.

The GDS-D was assessed in comparison to the original version (GDS-30), as well as the briefer versions (GDS-15, GDS-10, GDS-8, and GDS-4), using their respective cut-off scores (Li et al., 2015; Almeida and Almeida, 1999). To reflect real-world clinical scenarios, patients with comorbid depression and dementia were included in the depression group in comparison to the euthymic dementia cohort. Additionally, a subgroup analysis was conducted to compare the tools' ability to differentiate depression within the dementia cohorts.

2.6. Development of the GDS—D

The GDS-D was developed a priori, prior to any analysis of the study data, through a structured expert review process rather than a post-hoc, data-driven reduction. Clinicians with expertise across psychiatry, neurology, and geriatric medicine (J.F., P.J.N., E.A., B.L.) independently reviewed the items within the GDS-30 and identified items considered likely to be confounded by neurodegenerative disease processes, including symptoms related to cognitive impairment (Q14, Q26, Q29, Q30), apathy (Q2, Q4, Q12, Q20, Q28), and somatic symptoms (Q11, Q21). In addition, emotional and existential items that could be endorsed in the context of a diagnosis of a neurodegenerative dementia were excluded (Q1, Q5, Q8, Q10, Q13, Q15, Q19, Q22, Q23). For instance, in the context of a terminal illness, questions centred on future expectations, such as hope, may not necessarily indicate depression. This refinement resulted in the final 10-question version, the GDS—D. (Table 1).

2.7. Ethics approval

This study was approved by the Mater Misericordiae Ltd. Human Research Ethics Committee (EC00332), and a waiver of informed consent was granted in accordance with local ethical guidelines for the data retrospectively collected from patient electronic medical records.

3. Results

3.1. Diagnosing depression in a memory clinic

In differentiating depression ($n = 23$) from euthymic individuals with dementia ($n = 41$), the GDS-D exhibited the best overall accuracy compared with the other versions of the GDS. The overall diagnostic accuracy of the original GDS-30 was comparable to the briefer versions; however, the GDS-10 and GDS-4 performed worst with an accuracy of only 60–74 %. (Table 2).

3.2. Diagnosing depression in dementia

In assessing the utility of the GDS to differentiate depressed ($n = 18$) from euthymic patients with dementia ($n = 41$), the GDS-D also exhibited the best overall accuracy compared with the other versions of the GDS. The sensitivity of the GDS-D was similar to the GDS-30, but with improved specificity. (Supplementary Table 2). For the FTLD cohort, the GDS-D also had the best overall accuracy. Although the

Table 1
Development of the Geriatric Depression Scale, Dementias version (GDS—D).

GDS item	GDS-D	GDS-15	GDS-10	GDS-8	GDS-4
1. Are you basically satisfied with your life? [d]		+	+	+	+
2. Have you dropped many of your activities and interests? [a]		+	+		
3. Do you feel that your life is empty?	+	+	+	+	+
4. Do you often get bored? [d]		+		+	
5. Are you hopeful about the future? [d]					
6. Are you bothered by thoughts you can't get out of your head?	+				
7. Are you in good spirits most of the time?	+	+		+	
8. Are you afraid that something bad is going to happen to you? [d]		+	+		+
9. Do you feel happy most of the time?	+	+	+	+	+
10. Do you often feel helpless? [d]		+	+	+	
11. Do you often get restless and fidgety? [c]					
12. Do you prefer to stay at home, rather than going out and doing new things? [a]		+			
13. Do you frequently worry about the future? [d]					
14. Do you feel you have more problems with memory than most? [b]		+	+		
15. Do you think it is wonderful to be alive now? [d]		+		+	
16. Do you often feel downhearted and blue?	+				
17. Do you feel pretty worthless the way you are now?	+	+			
18. Do you worry a lot about the past?	+				
19. Do you find life very exciting? [d]					
20. Is it hard for you to get started on new projects? [a]					
21. Do you feel full of energy? [c]		+	+		
22. Do you feel that your situation is hopeless? [d]		+	+	+	
23. Do you think that most people are better off than you are?		+	+		
24. Do you frequently get upset over little things?	+				
25. Do you frequently feel like crying?	+				
26. Do you have trouble concentrating? [b]					
27. Do you enjoy getting up in the morning?	+				
28. Do you prefer to avoid social gatherings? [a]					
29. Is it easy for you to make decisions? [a]					
30. Is your mind as clear as it used to be? [b]					

[a] Apathy symptoms.
[b] Cognitive impairment symptoms.
[c] Somatic symptoms.
[d] Confounding emotional symptoms.

sensitivity of the GDS-D and GDS-30 was similar, the GDS-D demonstrated better specificity by not misidentifying euthymic FTLD as depressed. (Supplementary Table 3). For the AD cohort, the GDS-D once again had the best overall accuracy with a similar sensitivity to the GDS-30 but better specificity. (Supplementary Table 4).

4. Discussion

The GDS-D exhibited superior test performance to the GDS-30, and briefer GDS versions, in identifying depression in a memory clinic population when compared to the gold-standard of a psychiatric assessment and DSM-5 criteria. The exclusion of cognitive, apathy-related, and other potentially confounding items enhanced the diagnostic accuracy of the GDS-D relative to other abbreviated versions of the GDS. In addition, the GDS-D was slightly better than the GDS-30 and other versions in identifying depression in a dementia cohort of FTLD and AD. Overall, this preliminary study supports the utility of the GDS-D as a superior tool to diagnose depression in memory clinics.

The GDS-15 was developed as a shortened version of the GDS-30, intended to be more suitable for dementia cohorts (Yesavage and Sheikh, 1986). Its design, however, was based on selecting items with the highest correlation to depression in a psychiatric cohort, rather than focusing on the item's relevance to depression in a dementia cohort. Accordingly, the GDS-15 includes questions that could confound the diagnosis of depression in a patient with dementia. The even briefer versions of the GDS unfortunately also includes these questions. This may explain why these measures have been shown to lack accuracy in diagnosing depression in dementia cohorts (Goodarzi et al., 2017; Atchison et al., 2024). The GDS-D format proposed in this study avoided these confounds by removing questions that might be endorsed by a euthymic individual with dementia.

This study has some limitations related to its retrospective design and modest sample size. Post-hoc calculations based on the observed performance of the GDS-D suggest that, in the memory clinic comparison, approximately 56 individuals with depression and 36 euthymic dementia patients would be required to achieve ±5 % margin of error at a 95 % confidence level. While the present study exceeded the requirement for specificity, the depression-only cohort was underpowered. In the dementia-only subgroup, a future prospective study would require approximately 81 dementia patients (including 19 with comorbid depression) to achieve similar precision. The present cohort ($n = 59$, including 18 with comorbid depression) approached, but did not meet this target. Therefore, while the findings across the dementia subgroups support the GDS-D as a promising and clinically useful screening tool, sensitivity estimates should be interpreted with caution, and further validation in a larger cohort is warranted.

This study also has some other limitations related to patient inclusion and methodology. Patients with moderate to severe dementia, as defined by a MMSE of <15, were excluded due to the decreased reliability of the GDS in this population (n.b. the GDS relies on people being able to comprehend and respond to a series of questions). In more advanced dementia, care-giver rated tools have more clinical value (Brown et al., 2009). Although this study had a strong design by including patients with a high degree of diagnostic certainty, it nonetheless included retrospective data and a small sample of patients in which the GDS-30 was not applied on the same day as the psychiatric assessment. Future research should assess the GDS-D in larger prospective studies, undertake formal psychometric evaluation of internal validity and reliability, and evaluate the GDS-D across a wider range of neurodegenerative conditions.

5. Conclusion

This study showed that the GDS-D is a superior tool for diagnosing depression in a memory clinic setting. By removing potentially confounding questions, the GDS-D outperformed existing versions of the GDS.

CRediT authorship contribution statement

Joshua Flavell: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Kailas Roberts:** Writing – review & editing, Writing – original draft, Investigation.

Table 2
Test performance of the GDS in differentiating depression from *euthymic* dementia.

Test	Cut-off	Sensitivity	Specificity	Youden-index	PPV	NPV	Accuracy
GDS-D	≥4	82.6	97.6	80.2	95.0	90.5	92.2
GDS-30	≥10	91.3	78.0	69.4	70.0	94.1	82.9
GDS-15	≥6	78.3	87.8	66.1	78.3	87.8	84.4
GDS-10	≥5	65.2	95.1	60.3	88.2	83.0	60.3
GDS-8	≥4	47.8	100.0	47.8	100.0	77.4	81.2
GDS-4	≥1	69.6	75.6	45.2	61.5	81.6	73.4

PPV = Positive Predictive Value; NPV = Negative Predictive Value; GDS-D = Geriatric Depression Scale, Dementia version; GDS-30 = GDS, 30-item version; GDS-15 = GDS, 15-item version; GDS-10 = GDS, 10-item version; GDS-8 = GDS, 8-item version; GDS-4 = GDS, 4-item version.

Philip L.P. Morris: Writing – review & editing, Writing – original draft, Investigation. **Philip E. Mosley:** Writing – review & editing, Writing – original draft, Investigation. **Emily G.M. Ahern:** Writing – review & editing, Visualization, Investigation. **Benignus Logan:** Writing – review & editing, Validation, Investigation. **Robert J. Adam:** Writing – review & editing, Validation. **Caitlin A.T. McElligott:** Resources, Data curation. **Thomas B. Shaw:** Writing – review & editing, Visualization, Validation, Formal analysis. **Peter J. Nestor:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Research support

This study received financial support from the Mater Foundation and the Vonnie Healy Research Grant.

Relationships

There are no additional relationships to disclose.

Patents and intellectual property

There are no patents to disclose.

Other activities

There are no additional activities to disclose.

Declaration of Generative AI and AI-assisted technologies in the writing process

No generative AI or AI-assisted technologies were used in the writing or preparation of this manuscript.

Declaration of competing interest

There are no conflicts of interest to declare.

Acknowledgements

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2025.121056.

Data availability

The data that support the findings of this study are available upon reasonable request from the corresponding author. Data sharing is subject to ethical approval (Mater Misericordiae Ltd. Human Research Ethics Committee, EC00332).

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